CAMPUS SCHOLARSHIP AND MANAGEMENT PORTAL

A MAIN PROJECT REPORT

Submitted by

SANTHIYA K - (6176AC22UCS131)

SRINITHI S M - (6176AC22UCS146)

SURESH KUMAR S - (6176AC22UCS151)

in partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

ADHIYAMAAN COLLEGE OF ENGINEERING

(AN AUTONOMOUS INSTITUTION)

DR.M.G.R NAGAR, HOSUR-635130

ANNA UNIVERSITY: CHENNAI 600 025

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BONAFIDE CERTIFICATE

Certified that this main project report **"CAMPUS SCHOLARSHIP AND MANAGEMENT PORTAL"** is the Bonafide work of "**SANTHIYA K (6176AC22UCS131), SRINITHI SM (6176AC22UCS146), SURESH KUMAR S (6176AC22UCS151)"** who carried out the project under my supervision.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

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ABSTRACT

The Scholarship Management Portal is a web-based system developed to automate and streamline the scholarship administration process within a college campus. The system eliminates manual paperwork by providing a centralized digital platform for students, administrators, coordinators, and finance officers. It supports online scholarship applications, document uploads, multi-level verification, approval workflows, and fund disbursement tracking.

The portal is developed using modern web technologies including React for frontend, Node.js with Express for backend, and PostgreSQL/MySQL as the database. The system ensures role-based access control, secure authentication using JWT, and proper data validation. Automated notifications enhance transparency and communication between stakeholders. This solution improves efficiency, reduces processing time, enhances data security, and ensures accountability in scholarship management.

In addition to simplifying administrative operations, the system provides real-time monitoring and reporting features that assist management in making data-driven decisions. Dashboards and analytics help track the number of applications received, approved, rejected, and pending at various stages. This structured workflow minimizes delays and ensures that eligible students receive financial assistance in a timely manner.

Furthermore, the portal is designed with scalability and security as primary considerations, allowing it to handle increasing numbers of users and scholarship schemes in the future. The modular architecture enables easy maintenance and future enhancements such as mobile application integration, cloud deployment, and AI-based eligibility analysis. Overall, the system demonstrates an effective application of full-stack web development technologies to modernize scholarship management in educational institutions.

Keywords: Scholarship Management, Web-Based Application, Role-Based Access Control (RBAC), Online Application System, Multi-Level Approval Workflow, JWT Authentication.

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LIST OF ABBREVIATIONS

UI	–	User Interface
UX	–	User Experience
DB	–	Database
RDBMS	–	Relational Database Management System
SQL	–	Structured Query Language
JWT	–	JSON Web Token
API	–	Application Programming Interface
REST	–	Representational State Transfer
RBAC	–	Role-Based Access Control
HTTPS	–	Hyper Text Transfer Protocol Secure
CRUD	–	Create, Read, Update, Delete
IDE	–	Integrated Development Environment
OS	–	Operating System
JSON	–	JavaScript Object Notation

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CHAPTER 1
INTRODUCTION

1.1 OVERVIEW

Scholarships play a vital role in supporting students financially and promoting academic excellence. However, traditional scholarship management methods rely heavily on paperwork and manual processing, which often lead to delays, data mismanagement, and lack of transparency. The Scholarship Management Portal is developed to digitize this process and provide a secure and efficient web-based solution.

The system enables students to apply online, upload documents, and track application status. Administrators can create scholarship schemes, verify eligibility, and monitor applications. The finance department can manage fund disbursement and generate reports. This integrated platform ensures better coordination among all stakeholders.

In the conventional system, communication between students and administration is often slow and unclear. Students may not receive timely updates about their application status, and administrators face difficulties in managing large volumes of data manually. By introducing a centralized digital platform, the proposed system ensures real-time updates, quick communication, and improved transparency throughout the scholarship lifecycle.

The portal is designed with a user-friendly interface to ensure accessibility for all users regardless of their technical expertise. Role-based dashboards provide customized views for students, coordinators, administrators, and finance officers. This structured access control not only enhances security but also simplifies workflow management by clearly defining responsibilities at each stage of the process.

Moreover, the system is scalable and adaptable to future institutional requirements. It can support multiple scholarship schemes, varying eligibility criteria, and increasing numbers of users without affecting performance.

1.2 OBJECTIVE

The main objective of the **Scholarship Management Portal** is to automate and streamline the scholarship administration process within a college campus. The system aims to eliminate manual paperwork, reduce processing delays, improve transparency, and ensure secure and efficient management of scholarship applications from submission to fund disbursement.

Specific Objectives:

- **Automation of Scholarship Processing:**
  To automatically manage the complete scholarship workflow including application submission, document verification, approval processes, and fund disbursement tracking.

- **Centralized Data Management:**
  To maintain organized and secure records of students, scholarship schemes, eligibility criteria, applications, verification status, and payment details in a centralized database.

- **Role-Based Access Control:**
  To implement secure role-based authentication and authorization for students, administrators, coordinators, and finance officers to ensure controlled system access.

- **Online Application and Document Upload:**
  To enable students to apply for scholarships online and upload required documents digitally, reducing physical paperwork and manual submission.

- **Multi-Level Verification System:**
  To ensure applications pass through structured verification stages (department-level, admin-level, finance-level) before final approval.

CHAPTER 2
LITERATURE SURVEY

[1] Kumar, S., & Patel, R. presented "Web-Based Scholarship Management System for Educational Institutions."
This study focused on the development of a centralized online portal to manage scholarship applications within academic institutions. The authors highlighted the importance of digital record maintenance and automated verification to reduce administrative workload. The research emphasized improved transparency and efficiency through structured workflow management and secure login systems.

[2] Sharma, P., & Verma, A. presented "Design and Implementation of an Online Scholarship Application System."
This paper discussed the transition from manual scholarship processing to a web-based solution. The authors demonstrated how database-driven systems can manage large volumes of student data while maintaining accuracy. The research concluded that automation significantly reduces delays and minimizes documentation errors.

[3] Lee, H., & Park, J. presented "Role-Based Access Control in Academic Management Systems."
The study examined the implementation of RBAC models in multi-user academic portals. The authors emphasized enhanced data security and restricted access control as critical factors in protecting sensitive student information. The findings showed that structured access levels improve accountability and system reliability.

[4] Ahmed, M., & Khan, S. presented "Digital Scholarship Processing Using Web Technologies."
This research explored the integration of frontend and backend technologies to build scalable scholarship portals. The authors highlighted RESTful APIs and secure authentication mechanisms as essential components for system performance and data integrity.

[5] Rao, V., & Mehta, N. presented "Database-Driven Financial Aid Management System."
The paper analyzed relational database design for managing student financial records. It emphasized normalization techniques, indexing strategies, and efficient query handling to ensure fast retrieval of scholarship and payment information.

[6] Singh, A., & Kaur, M. presented "Secure Document Verification System for Online Applications."
This study addressed digital document upload and verification processes in academic portals. The authors proposed file validation mechanisms and multi-stage approval workflows to ensure authenticity and prevent fraudulent submissions.

[7] Chen, Y., & Liu, X. presented "Workflow Automation in Educational Administration Systems."
The research focused on automating administrative processes through structured approval pipelines. The authors demonstrated that multi-level approval systems reduce human errors and enhance monitoring of application progress.

[8] Thomas, D., & Wilson, P. presented "Implementation of JWT-Based Authentication in Web Applications."
This paper discussed secure authentication methods using JSON Web Tokens. The authors highlighted stateless session management, enhanced security, and efficient authorization in web-based systems handling sensitive information.

[9] Martinez, L., & Brown, T. presented "Cloud-Based Academic Portal Systems."
The study examined scalable web architectures for educational institutions. The authors emphasized modular system design and cloud deployment strategies to support growing numbers of users and services.

[10] Gupta, R., & Nair, P. presented "Automation and Transparency in Scholarship Distribution Systems."
This research analysed the impact of digital portals on improving transparency in scholarship allocation. The authors concluded that automated notifications, digital records, and centralized monitoring significantly enhance accountability and operational efficiency.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

The existing scholarship management process in many colleges is primarily manual and paper-based. Students are required to collect physical application forms from the administrative office, fill them manually, and submit them along with photocopies of required documents such as mark sheets, income certificates, identity proofs, and bank details. This process consumes significant time and effort for both students and administrative staff.

In the manual system, verification is carried out by different departments through physical file movement. Applications are passed from department coordinators to administrative officers and then to the finance department for approval and fund disbursement. This sequential process often leads to delays, misplaced documents, and lack of real-time tracking.

Record maintenance in the existing system is typically handled through physical registers or basic spreadsheet files. Such methods increase the risk of data duplication, human errors, data loss, and inconsistencies. Retrieving past records for audit or reporting purposes becomes difficult and time-consuming.

Several limitations are evident in the existing approach

- Manual paperwork leads to excessive documentation.
- High probability of human errors and data duplication.
- No real-time status tracking for students.
- Risk of document loss or damage.
- Poor scalability when the number of applications increases.

3.2 PROPOSED SYSTEM

The proposed Scholarship Management Portal is a web-based centralized system designed to automate and digitize the entire scholarship process within the college campus. The system provides an integrated platform where students, administrators, coordinators, and finance officers can perform their tasks efficiently.

Students can register, log in securely, apply for scholarships online, upload required documents, and track their application status in real time. The system eliminates physical paperwork by storing all data digitally in a secure database.

The proposed system introduces a structured multi-level verification workflow. Applications are verified digitally by department coordinators, approved by administrators, and processed by the finance department for fund disbursement. Each stage is recorded in the system, ensuring accountability and transparency.

The portal uses modern web technologies such as React (frontend), Node.js with Express (backend), and PostgreSQL/MySQL (database). It implements role-based access control (RBAC) and JWT-based authentication to ensure data security and restricted access.

Automated notifications (email/system alerts) inform students about application submission, verification updates, approval, rejection, or required corrections. The system also supports report generation, scholarship analytics, and fund tracking.

Advantages of Proposed System

- Eliminates paperwork and manual errors.
- Reduces scholarship processing time.
- Ensures secure and structured data management.
- Improves transparency and accountability.
- Enhances user experience through an intuitive interface.
- Provides real-time monitoring and analytics.
- Easily adaptable for future expansion.

3.3 PROPOSED SOLUTION

The proposed solution for the **Scholarship Management Portal** is to design and implement a secure, web-based automated system that digitizes and streamlines the entire scholarship lifecycle within a college campus. The solution eliminates manual paperwork and fragmented record-keeping by introducing a centralized platform capable of managing scholarship applications, verification workflows, approval processes, and fund disbursement in a structured and transparent manner.

The system accepts various inputs such as student registration details, academic records, income information, scholarship eligibility criteria, and required supporting documents. These inputs are validated and processed through a multi-level verification workflow involving department coordinators, administrators, and finance officers. Each stage of verification updates the application status in real time.

The backend system applies business logic rules to evaluate eligibility conditions such as academic performance thresholds, income limits, attendance requirements, and document completeness. Applications that satisfy all mandatory criteria proceed to the approval stage, while incomplete or ineligible applications are flagged for correction or rejection. The system maintains an audit trail to record every action taken by authorized users.

By automating this process, the proposed solution ensures that

- No duplicate or fraudulent applications are submitted.
- Each application is verified through a structured multi-level approval process.
- Eligibility criteria are strictly validated according to scholarship rules.
- Students can track their application status in real time.
- All scholarship data is stored securely in a centralized database.

3.4 IDEATION & BRAINSTORMING

In the context of our Scholarship Management Portal project, the ideation and brainstorming phase focused on identifying innovative ways to digitize scholarship administration, reduce manual intervention, and ensure a transparent and secure approval process. Ideation involved exploring individual creative solutions for simplifying application handling, while brainstorming sessions as a team helped refine and prioritize the most practical, scalable, and secure ideas for implementation.

Online Scholarship Application System

A fully web-based application system is proposed to allow students to register, apply for scholarships, and upload required documents online. The system validates input data and ensures that all mandatory fields and documents are submitted before final application. This significantly reduces paperwork and eliminates manual form handling errors.

Multi-Level Verification and Approval Workflow

The system includes a structured multi-stage verification process involving department coordinators, administrators, and finance officers. Each application moves systematically through verification, approval, and fund disbursement stages. The workflow ensures accountability and prevents unauthorized approvals while maintaining a clear audit trail.

Eligibility Validation Mechanism

An automated eligibility checking mechanism is integrated into the system. It verifies academic performance, income criteria, attendance records, and other scholarship requirements before forwarding applications for approval. This ensures that only eligible students proceed to the next stage, reducing administrative burden.

Role-Based Access Control and Security

The portal implements role-based authentication to provide secure access to different users such as students, coordinators, administrators, and finance officers. Secure login using JWT authentication and encrypted data handling protects sensitive student and financial information.

3.5 PROBLEM SOLUTION FIT

In the context of our **Scholarship Management Portal**, achieving a Problem–Solution Fit is essential. It ensures that the system being developed directly addresses the real challenges faced by educational institutions in managing scholarship programs. This alignment between the identified problems and the proposed digital solution improves operational efficiency, reduces administrative burden, enhances transparency, and ensures secure handling of sensitive student data.

By carefully analyzing institutional needs, student expectations, and administrative workflows, the system provides an automated, structured, and user-friendly scholarship management experience. It streamlines the entire lifecycle—from application submission to fund disbursement—while maintaining accountability and compliance with institutional policies. The primary focus is to build a scalable, secure, and reliable scholarship management system that delivers practical value and convenience to both students and administrators.

1. Customer Segment(s)

Educational institutions such as colleges and universities that manage multiple scholarship schemes and require efficient administration systems. Administrative staff and scholarship coordinators responsible for verifying applications, approving eligible candidates, and maintaining scholarship records.

Finance officers who handle fund allocation and disbursement tracking. Students who apply for scholarships and require a transparent, easy-to-use platform to submit applications and track their status.

These users benefit from a centralized system that automates application handling, ensures secure data storage, and reduces manual workload while improving processing speed and accuracy.

2. Jobs to Be Done / Problems

Time-consuming application processing due to physical form submission and document verification. High risk of human errors in eligibility checking and record maintenance. Lack of real-time status tracking, causing repeated inquiries from students.

Misplacement or damage of physical documents. Difficulty in managing multiple scholarship schemes simultaneously. Limited transparency in approval and rejection decisions. Challenges in generating reports and tracking fund utilization.

Difficulty in ensuring data privacy and preventing unauthorized access to sensitive student information. Inefficient communication between departments during multi-level verification and approval stages.

Inability to maintain proper audit trails for monitoring actions taken on each application.

The proposed Scholarship Management Portal directly addresses these problems by automating workflows, implementing multi-level verification, providing real-time tracking, and maintaining secure digital records. This ensures that the solution effectively fits the real needs of the institution and its stakeholders.

3.6 ARCHITECTURE DESIGN

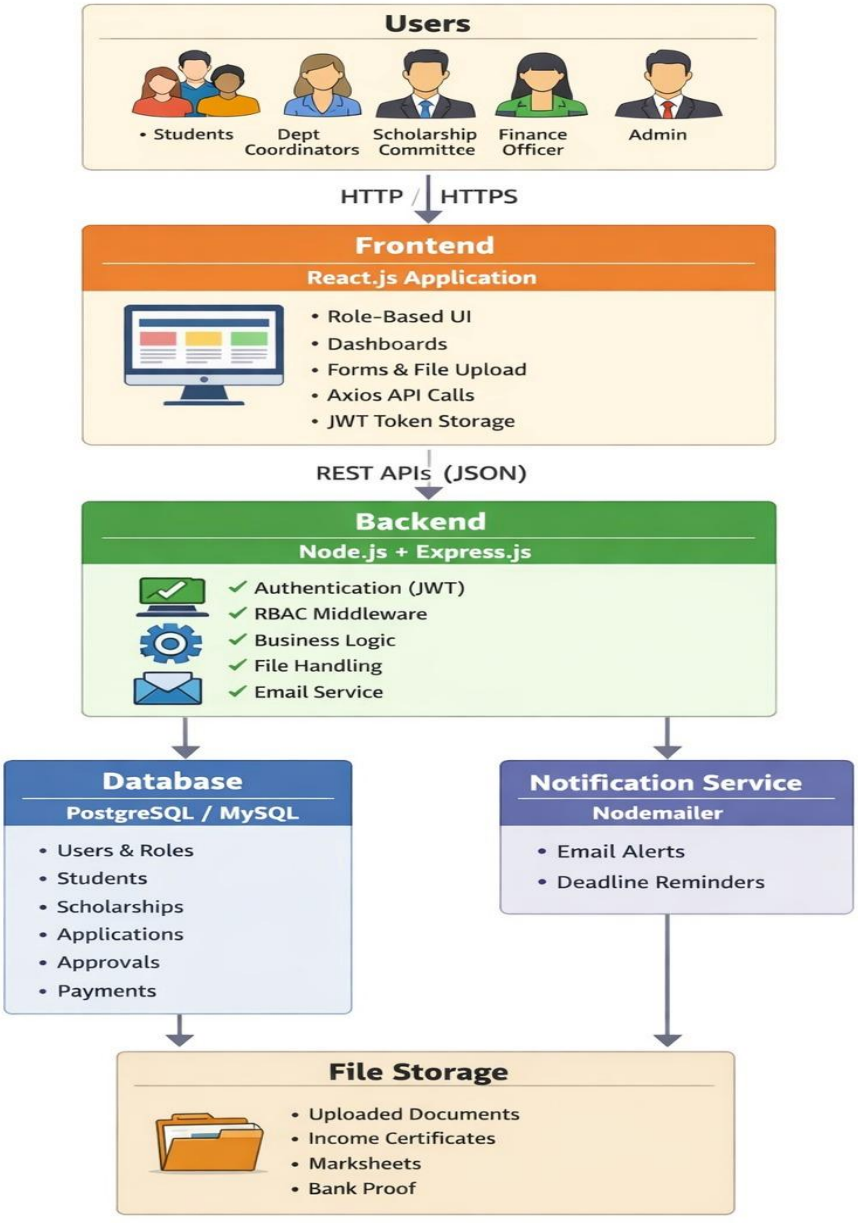


Figure 3.1: Model Architecture

1. Start and User Access

Start:

The process begins when a user opens the Scholarship Management Portal through a web browser.

User Accesses the System:

Different types of users (Students, Department Coordinators, Scholarship Committee Members, Finance Officers, and Admin) access the system via HTTP/HTTPS.

2. **Login and Authentication:**

   The user enters login credentials. The system verifies the credentials using JWT-based authentication.

- If valid → Access is granted based on role.
- If invalid → User is prompted to re-enter credentials.

3. **Frontend Interaction Phase (React.js Application)**

   This phase manages user interaction through a role-based interface.

   a) Role-Based Dashboard Access

   After successful login:

- Students access the scholarship application dashboard.
- Coordinators access verification panels.
- Admin accesses management dashboard.
- Finance Officer accesses payment management section.

   b) Form Submission and File Upload

   Students:

- Fill scholarship application forms.
- Upload required documents (income certificate, mark sheets, bank proof).
- Submit applications digitally.

   The frontend sends data to the backend using REST APIs (JSON format) via Axios.

   c) JWT Token Handling

- JWT token is stored securely.

- Token is attached to API requests for secure communication.

4. **Backend Processing Phase (Node.js + Express.js)**

   This phase handles server-side logic and validation.

   a) Authentication & Authorization

   - JWT token is verified.
   - RBAC (Role-Based Access Control) middleware ensures users access only permitted resources.

   b) Business Logic Execution

   The system processes:

   - Scholarship eligibility validation.
   - Multi-level verification workflow.
   - Approval and rejection decisions.
   - Payment tracking logic.

   c) File Handling

   Uploaded documents are processed and stored securely in the file storage system.

   d) Email Notification Service

   The backend triggers email notifications using Nodemailer for:

   - Application submission confirmation.
   - Approval/rejection updates.
   - Deadline reminders.

5. Database Management Phase (PostgreSQL / MySQL)

This phase ensures centralized data storage.

a) Data Storage Includes:

- Users & Roles
- Student Records
- Scholarship Schemes
- Applications
- Approval Status
- Payment Details

b) Data Retrieval & Updates

- Application status updates are stored.
- Approval records are maintained.
- Payment confirmations are recorded.

All operations ensure data consistency and integrity.

6. Multi-Level Verification Workflow

a) Department Coordinator Review

- Reviews application details and documents.
- Approves or forwards to next level.

b) Scholarship Committee Approval

- Final eligibility validation.
- Decision recorded in database.

c) Finance Officer Processing

- Initiates fund disbursement.
- Updates payment status.

Decision – Application Approved?

Yes → Payment processed and notification sent.

No → Rejection notification sent to student.

7. **Notification & Communication Phase**

Notification Service (Nodemailer):

- Sends email alerts.
- Sends deadline reminders.
- Sends approval/rejection confirmation.

This ensures transparency and real-time communication.

8. **End Process**

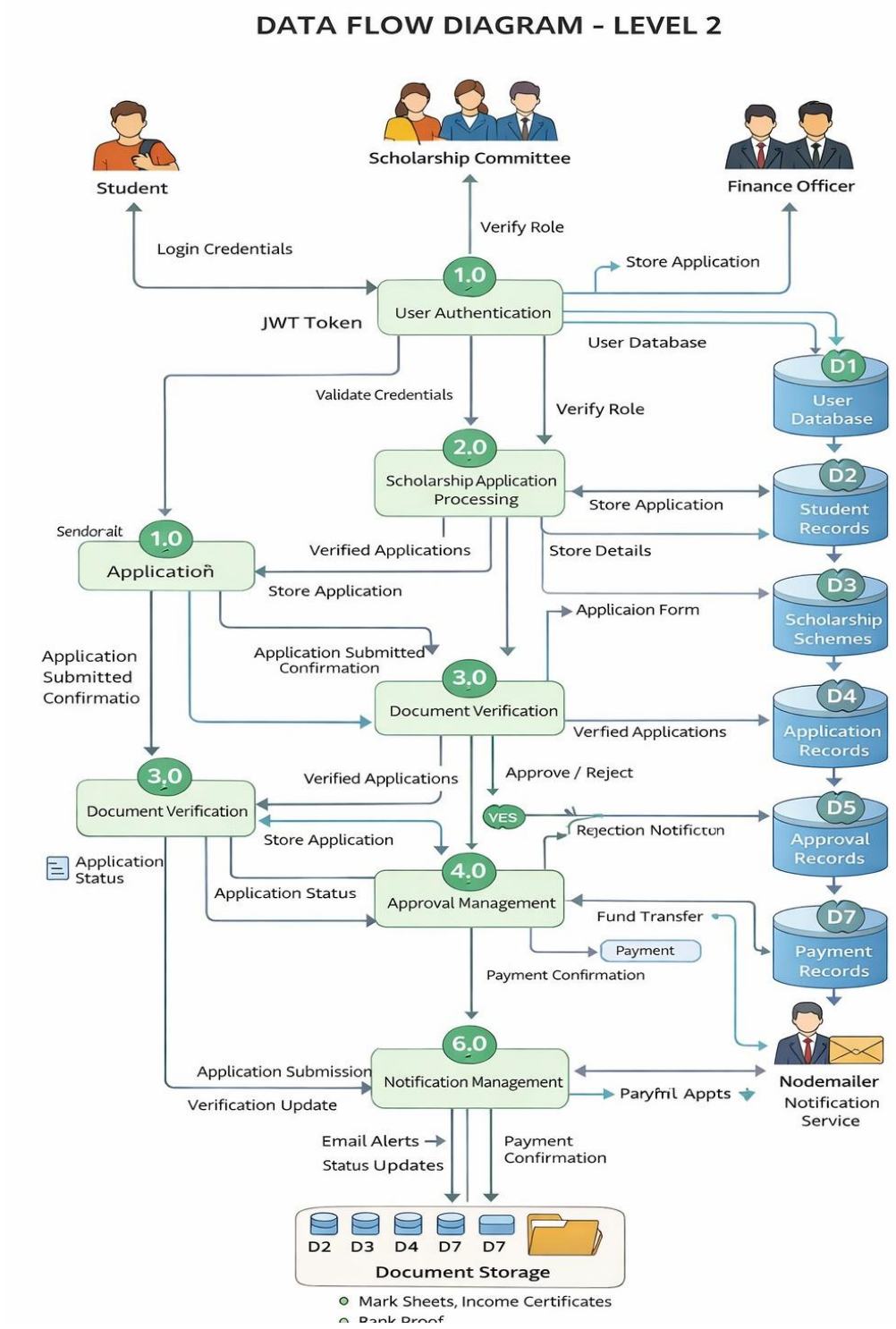
Student can:

- Track real-time application status.
- View approval/rejection details.
- Receive payment confirmation.

The system ensures:

- Secure processing
- Transparent workflow
- Centralized data management
- Reduced manual workload

3.7 DATA FLOW DIAGRAMS



The **Campus Scholarship Management Portal** follows a structured data flow architecture that enables efficient management of scholarship applications, verification processes, approvals, and fund disbursement. The system integrates a user-friendly frontend interface, a backend server for processing business logic, and a centralized database for storing scholarship-related information

Figure 3.2: Data Flow Diagram

Major Entities

- **Student** – Represents students who apply for scholarships by submitting their personal details, academic records, income certificates, and other required documents through the portal.
- **Department Coordinator** – Responsible for verifying student applications at the department level and ensuring that submitted details and documents are correct.
- **Scholarship Committee** – A group responsible for evaluating verified applications and approving or rejecting them based on eligibility criteria.
- **Finance Officer** – Responsible for processing scholarship payments and managing the disbursement of approved funds to students.
- **Administrator** – Manages the overall system including scholarship schemes, user roles, application records, and system monitoring.
- **Scholarship** – Represents various scholarship schemes offered by the institution, each with its own eligibility criteria, application deadlines, and funding limits.
- **Application** – Represents the scholarship application submitted by students along with supporting documents and approval status.

Data Flow Sequence

Step 1: Input Phase (Application Submission)

- The student logs into the Scholarship Management Portal through the web-based frontend interface. The student fills out the scholarship application form by entering personal information, academic details, income information, and uploading required documents such as income certificates, mark sheets, and bank details.

- The frontend performs initial validation to ensure that all mandatory fields are filled and the uploaded files meet format requirements. The application data is then transmitted to the backend server using HTTP POST requests. The backend validates the data and stores it securely in the database.

Step 2: Processing Phase (Verification and Approval)

- Once the application is submitted, the system forwards it to the Department Coordinator for initial verification. The coordinator checks the authenticity of student information and supporting documents. If the details are correct, the application is forwarded to the Scholarship Committee for further evaluation.
- The Scholarship Committee reviews the eligibility criteria such as academic performance, financial background, and scholarship availability. Based on the evaluation, the committee can approve or reject the application. The system automatically updates the application status and records all actions taken during the verification and approval process.

Step 2: Processing Phase (Verification and Approval)

- Once the application is submitted, the system forwards it to the **Department Coordinator** for initial verification. The coordinator checks the authenticity of student information and supporting documents. If the details are correct, the application is forwarded to the **Scholarship Committee** for further evaluation.
- The Scholarship Committee reviews the eligibility criteria such as academic performance, financial background, and scholarship availability. Based on the evaluation, the committee can approve or reject the application. The system automatically updates the application status and records all actions taken during the verification and approval process.

Step 4: Notification and Fund Disbursement

- After the scholarship application is approved by the committee, the **Finance Officer** processes the scholarship payment. The system records payment details and updates the application status to indicate successful disbursement.
- The notification service sends automated **email alerts or portal notifications** to students informing them about application submission, approval status, rejection,

or payment confirmation. This improves transparency and keeps all stakeholders informed throughout the scholarship lifecycle.

Flow Characteristics

Direction:

The data flow is **bidirectional**, where user inputs are sent to the backend for processing and the processed results such as application status and notifications are returned to the user interface.

Control Mechanism:

The system follows a **client-server architecture** that separates the frontend interface, backend logic, and database management for better scalability and maintainability.

Data Storage:

All scholarship-related data including applications, documents, approvals, and payment records are stored in a **centralized relational database** for secure access and long-term storage.

Validation:

Data validation is implemented at both the frontend and backend levels to ensure correctness, prevent duplicate applications, and maintain accurate scholarship records.

Result

The **Campus Scholarship Management Portal** efficiently manages the entire scholarship lifecycle from application submission to fund disbursement. The system ensures transparency, reduces administrative workload, and prevents manual errors. By organizing relationships between **students, scholarships, applications, approvals, and payments**, the portal provides a reliable and automated platform for scholarship administration within the campus.

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 HARDWARE REQUIREMENT

- Processor: Quad-core, 2.0 GHz or higher
- RAM: Minimum 4 GB (Recommended 8 GB for smooth execution)
- Hard Disk: Minimum 20 GB of free space
- Display: 1366 × 768 resolution or higher
- Input Devices: Standard Keyboard and Mouse
- Network: Internet connection for web access

4.2 SOFTWARE REQUIREMENTS

- Operating System: Windows 10 / 11 (64-bit) or Linux (Ubuntu 20.04 or later)
- Programming Language: Python (version 3.10 or above)
- Framework: React.js
- Frontend Technologies: HTML5, CSS3, JavaScript
- Database: PostgreSQL
- IDE / Code Editor: Visual Studio Code
- Browser: Google Chrome, Microsoft Edge, or Mozilla Firefox

CHAPTER 5

IMPLEMENTATION

5.1 DATA COLLECTION

The **Campus Scholarship and Management Portal** collects, processes, and manages student scholarship information through a centralized digital platform. The system is designed to streamline the process of scholarship application, verification, approval, and fund distribution within educational institutions. It gathers information from multiple stakeholders including students, administrators, scholarship providers, and academic departments.

The portal ensures that all scholarship-related data is organized, validated, and stored securely for efficient processing. By digitizing the collection of scholarship applications and academic details, the system minimizes manual paperwork and reduces processing time. The data collected helps administrators evaluate eligibility criteria, verify academic records, and allocate scholarships appropriately.

Data Input Sources

Student Input

Students are the primary data providers in the scholarship portal. They submit scholarship applications through the portal by entering required personal and academic information such as:

- Student personal details (name, ID, contact information)
- Academic details (department, semester, CGPA/percentage)
- Family income details
- Scholarship category or scheme selected
- Required supporting documents (income certificate, mark sheets, ID proof)

These inputs are submitted through secure web forms and stored in the system for verification and further processing.

Administrator Input

The administrator or scholarship coordinator manages the system and provides foundational data required for scholarship processing, including:

- Scholarship scheme details (name, eligibility criteria, amount)
- Department and course information
- Application deadlines and approval workflows
- Verification status updates
- Approval or rejection decisions for applications

Administrators also monitor and manage all student applications through the portal.

System Configuration Data

The backend framework automatically records system-generated metadata such as:

- Unique application IDs
- Submission timestamps
- Application status updates
- User authentication and login records

These configurations ensure that every application is uniquely identifiable and traceable throughout the scholarship lifecycle.

Data Processing and Validation

Before storing the information in the database, the system performs several validation checks to ensure data accuracy and integrity.

- Ensures that each student submits only one application per scholarship scheme.
- Verifies that all mandatory fields and documents are uploaded.
- Confirms that the student meets eligibility requirements such as minimum academic performance or income limit.
- Prevents duplicate student records or scholarship entries.

- Validates relationships between students, departments, and scholarship schemes.

The backend framework automatically validates data types, constraints, and relationships before committing records to the database.

Database Storage

All validated data are stored in the central relational database of the system. The database consists of several interconnected tables that maintain scholarship-related information.

- Student Table: Stores student personal and academic information.
- Scholarship Table: Contains details of scholarship schemes, eligibility criteria, and benefits.
- Application Table: Records each scholarship application submitted by students.
- Department Table: Maintains department details and their associated students.
- Document Table: Stores uploaded supporting documents submitted by applicants.
- Approval Table: Maintains verification and approval status for each application.

These interconnected tables allow the system to efficiently retrieve and manage scholarship information.

System-Generated Data

During the operation of the portal, the system automatically generates additional data to support scholarship management.

- **Application Status Records:** Tracks whether an application is pending, verified, approved, or rejected.
- **Notification Logs:** Records messages or alerts sent to students regarding application status updates.
- **Transaction Records:** Maintains details of scholarship fund disbursement to approved

students.

- **Audit Logs:** Stores records of administrative actions for transparency and accountability.

Data Flow During Operation

The data flow within the system follows a structured process.

- Students submit scholarship applications through the web interface.
- The application data is transmitted to the backend via HTTP POST requests.
- The system validates the submitted information and verifies uploaded documents.
- Once validated, the data is stored in the central database. Administrators review and verify the applications through the management dashboard.
- Based on eligibility criteria, the system updates the application status and approves or rejects the request.
- The final results and notifications are displayed to students through the portal interface.

Data Update and Maintenance

The system allows administrators to update scholarship schemes, modify eligibility criteria, or manage application records at any time. Students can also update their profiles and application details before the submission deadline.

All modifications are automatically reflected in the database to maintain consistency across the system. Administrative tools enable authorized users to review, edit, and manage collected data securely.

Algorithm Implementation

The scholarship portal uses rule-based eligibility evaluation algorithms to determine whether applicants qualify for a specific scholarship scheme.

The algorithm checks parameters such as academic performance, income criteria, department eligibility, and application completeness. Based on these constraints, the system automatically filters eligible applicants and presents them to administrators.

5.2 COMPONENTS DESIGN

The Campus Scholarship and Management Portal follows a modern three-tier architecture consisting of a frontend layer, backend layer, and database layer, along with supporting services such as notification and file storage. This architecture ensures scalability, security, and efficient communication between system components.

The system allows different types of users—including students, department coordinators, scholarship committee members, finance officers, and administrators—to interact with the portal through a web-based interface. Each user role is provided with role-based access to specific features of the system.

1. **Users Layer**

The top layer of the system consists of different users who interact with the portal through web browsers.

Types of Users

- Students
  Students register in the system, apply for scholarships, upload required documents, and track the status of their applications.
- Department Coordinators
  Department coordinators verify student details, validate academic records, and forward applications for further review.
- Scholarship Committee
  The scholarship committee evaluates verified applications and decides whether the student qualifies for the scholarship based on eligibility criteria.
- Finance Officer
  Finance officers manage the financial aspects of scholarships, including fund allocation and payment processing for approved students.
- Administrator (Admin)
  Administrators oversee the entire system, manage user roles, configure

scholarship schemes, and monitor all portal activities.

- Users interact with the system through HTTP/HTTPS requests, ensuring secure communication.

2. **Frontend Layer (React.js Application)**

The frontend layer is developed using React.js, which provides a responsive and interactive user interface. It serves as the primary interaction point between users and the system.

The frontend is responsible for collecting user inputs, displaying data, and communicating with backend services through API calls.

Key Features of the Frontend

- **Role-Based User Interface**
  Different users see different dashboards and functionalities based on their roles such as student, coordinator, or admin.
- **Dashboard System**
  Each user role has a customized dashboard displaying relevant information such as applications submitted, approvals pending, and scholarship status.
- **Application Forms and File Upload**
  Students can fill scholarship application forms and upload required documents such as income certificates, mark sheets, and bank details.
- **API Communication using Axios**
  The frontend uses Axios to send HTTP requests to backend REST APIs for data retrieval and submission.
- **JWT Token Storage**
  User authentication tokens are stored securely to maintain session management and ensure secure access to system resources.

3. Backend Layer (Node.js + Express.js)

The backend layer is implemented using Node.js with Express.js**,** which acts as the core processing unit of the system. It handles business logic, authentication, and communication with the database.

The backend receives requests from the React frontend, processes the data, and returns responses in JSON format through REST APIs**.**

Major Backend Components

- **Authentication System (JWT)**

  The system uses **JSON Web Token (JWT)** authentication to verify user identity and maintain secure login sessions.

- **Role-Based Access Control (RBAC)**

  RBAC middleware ensures that users can only access features permitted for their specific roles.

- **Business Logic Processing**

  The backend processes scholarship applications, validates eligibility conditions, manages approvals, and performs necessary operations related to scholarship management.

- **File Handling Module**

  Handles uploading and storing student documents such as income certificates, academic records, and bank details.

- **Email Notification Service**

  The backend integrates with **Nodemailer** to send automated email notifications regarding application submissions, approvals, and reminders.

4. Database Design (SQLite)

The system uses a **relational database such as PostgreSQL or MySQL** to store all scholarship-related data securely.

The database maintains structured tables that store information about users, scholarship schemes, applications, approvals, and financial transactions.

Major Database Tables

- **Users and Roles Table**
  Stores user login credentials and role information for authentication and authorization.

- **Students Table**
  Contains student personal details, academic information, and department association.

- **Scholarships Table**
  Stores details of scholarship schemes including eligibility criteria, scholarship amount, and deadlines.

- **Applications Table**
  Records scholarship applications submitted by students.

- **Approvals Table**
  Tracks verification status and approval decisions made by coordinators or committee members.

- **Payments Table**
  Maintains records of scholarship fund disbursements processed by finance officers.

5. Notification Service (Nodemailer)

The system integrates Nodemailer to provide automated email notifications for important events.

Notification Features

- **Email Alerts**
  Students receive notifications when their scholarship applications are submitted, reviewed, approved, or rejected.
- **Deadline Reminders**
  The system sends reminders to students before scholarship application deadlines to encourage timely submissions.

6. File Storage System

The system includes a **file storage component** that securely stores documents uploaded by students during the scholarship application process.

Stored Documents

- Income certificates
- Academic mark sheets
- Identity proofs
- Bank account proof documents

These files are linked to student applications and retrieved when verification is required.

Architecture Workflow

The overall workflow of the system operates as follows:

1. Users access the system through the React frontend interface.
2. User requests are sent to the Node.js backend through REST APIs.
3. The backend processes requests using business logic and RBAC security mechanisms.
4. Data is stored or retrieved from the PostgreSQL/MySQL database.
5. Uploaded documents are stored in the file storage system.

5.3 SOFTWARE DESCRIPTION

React.js

React.js is a popular JavaScript library used for building interactive and dynamic user interfaces. It is used as the primary frontend technology for the Campus Scholarship and Management Portal. React enables the development of reusable UI components and provides a fast and responsive user experience through its virtual DOM mechanism. In this project, React is used to create role-based dashboards, scholarship application forms, and various interactive components that allow users such as students, administrators, and finance officers to interact efficiently with the system.

Node.js

Node.js is a server-side JavaScript runtime environment built on Chrome's V8 engine. It enables developers to build scalable and high-performance backend applications. In the Campus Scholarship and Management Portal, Node.js handles server-side operations such as processing scholarship applications, managing user authentication, handling document uploads, and communicating with the database. Its event-driven architecture ensures efficient handling of multiple user requests simultaneously.

Express.js Framework

Express.js is a lightweight web application framework for Node.js that simplifies backend development. It provides a robust set of features for building RESTful APIs and handling HTTP requests and responses. In this project, Express.js manages routing, middleware functions, authentication mechanisms, and business logic related to scholarship management. It acts as the bridge between the React frontend and the database, ensuring smooth data flow across system components.

PostgreSQL / MySQL Database

PostgreSQL or MySQL serves as the relational database management system for the Campus Scholarship and Management Portal. The database stores all structured data including student profiles, scholarship schemes, application records, approval status, and payment details. It ensures data integrity, efficient querying, and reliable storage of large volumes of scholarship-related data. The relational structure allows different entities such as students, scholarships, and applications to be interconnected and easily managed.

JWT Authentication

JSON Web Token (JWT) is used for secure user authentication and session management in the system. After successful login, the server generates a token that is stored on the client side and used for subsequent authenticated API requests. JWT ensures secure communication between the frontend and backend by verifying the identity of users before granting access to protected resources.

Axios (API Communication)

Axios is a promise-based HTTP client used in the React frontend to communicate with the backend APIs. It enables the application to send GET, POST, PUT, and DELETE requests to the Node.js server. In the Campus Scholarship and Management Portal, Axios is used for tasks such as submitting scholarship applications, retrieving application status, and managing user data.

Nodemailer

Nodemailer is a Node.js module used for sending email notifications from the application. It plays a crucial role in the Campus Scholarship and Management Portal by sending automated emails related to scholarship applications. Students receive notifications when applications are submitted, approved, rejected, or when deadlines are approaching.

5.4 RESULT

The Campus Scholarship and Management Portal was successfully developed and implemented as a web-based application that automates and simplifies the scholarship management process within an educational institution. The system integrates multiple modules including user authentication, scholarship application management, document upload, approval workflow, and notification services. By utilizing modern web technologies such as React.js for the frontend, Node.js with Express.js for the backend, and PostgreSQL/MySQL for the database, the system provides a scalable, secure, and efficient platform for managing scholarship-related activities.

The portal allows students to register, log in securely, and submit scholarship applications through a user-friendly interface. Students can upload required documents such as income certificates, mark sheets, and bank details directly through the system. This eliminates the need for manual paperwork and physical submission of documents. The system stores all uploaded files securely and maintains organized digital records for easy access and verification.

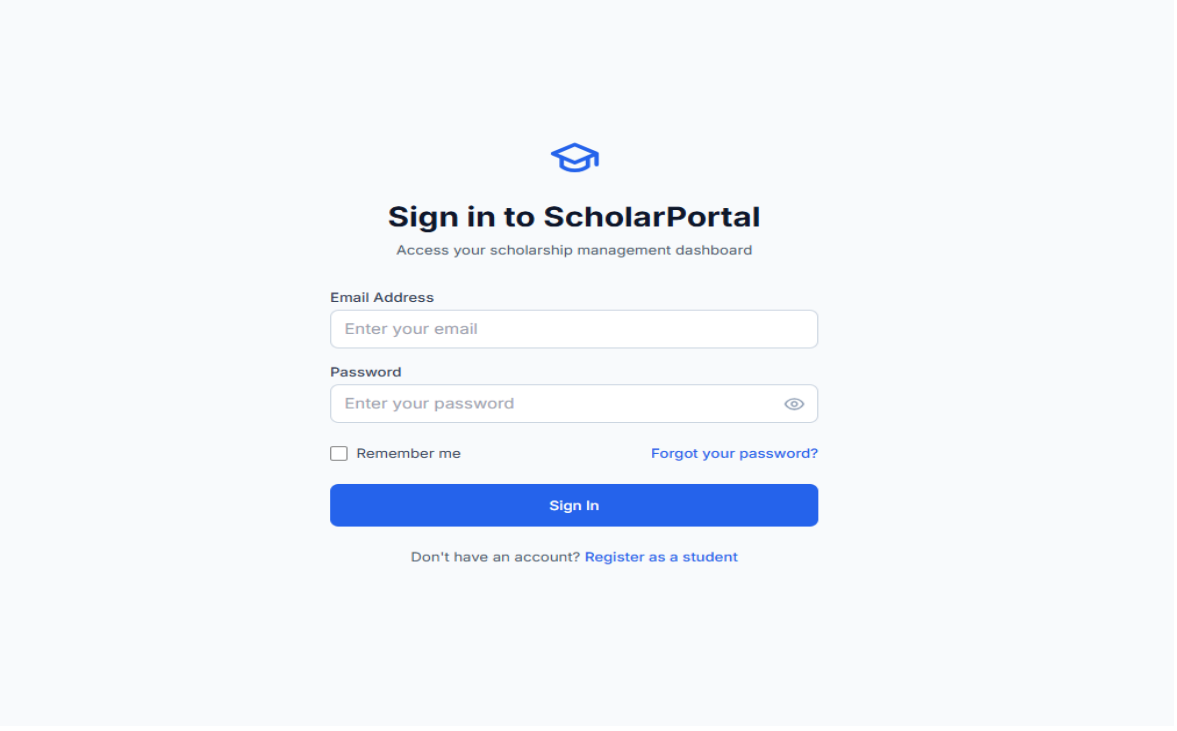


Figure 5.1: login Page

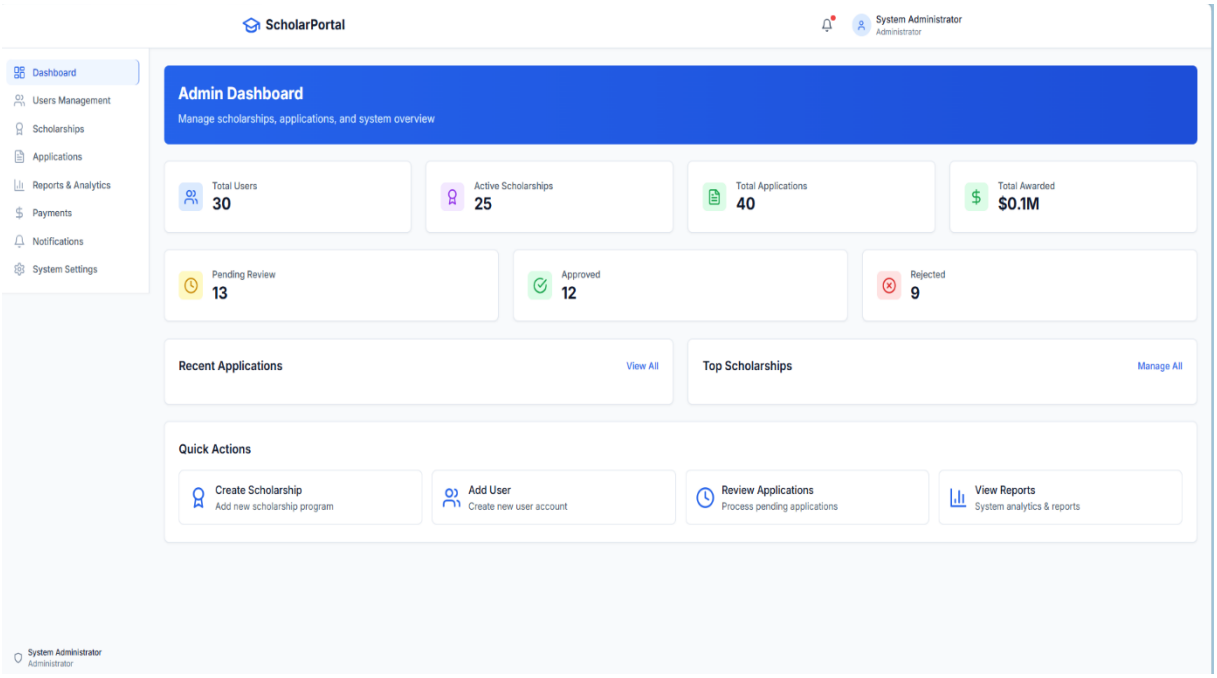


Figure 5.2: Admin Dashboard

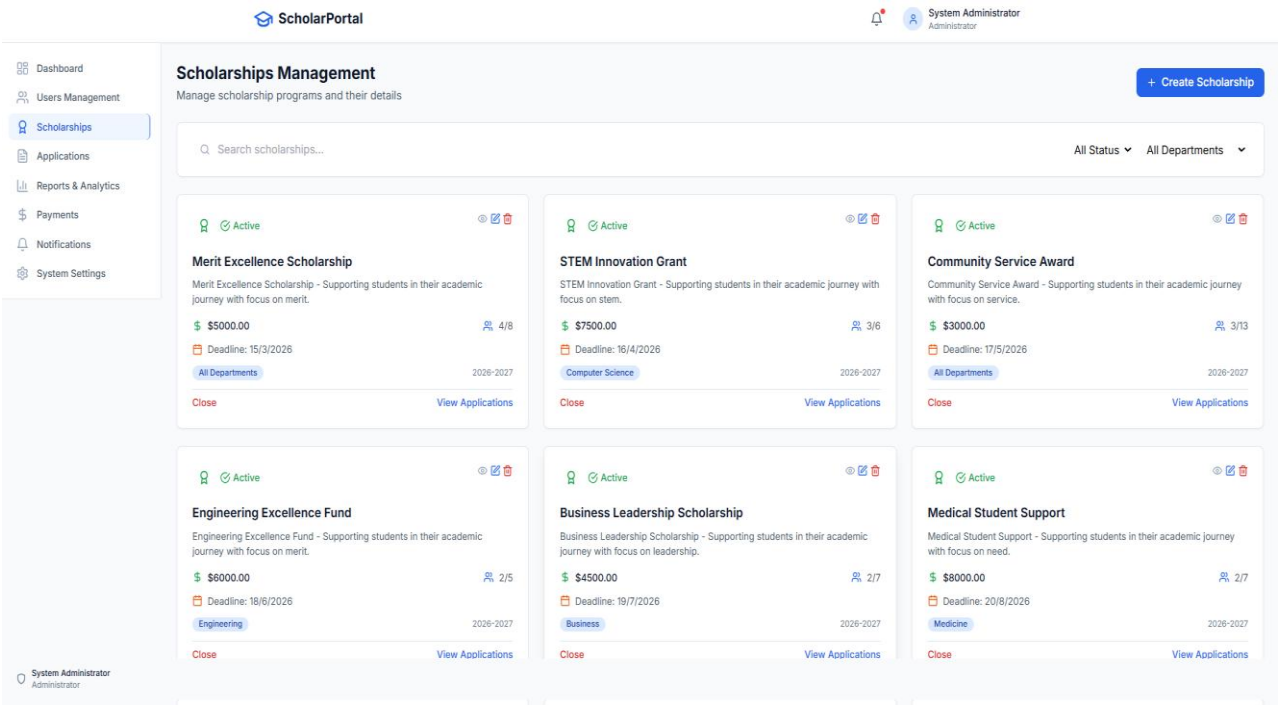


Figure 5.3: Scholarship page

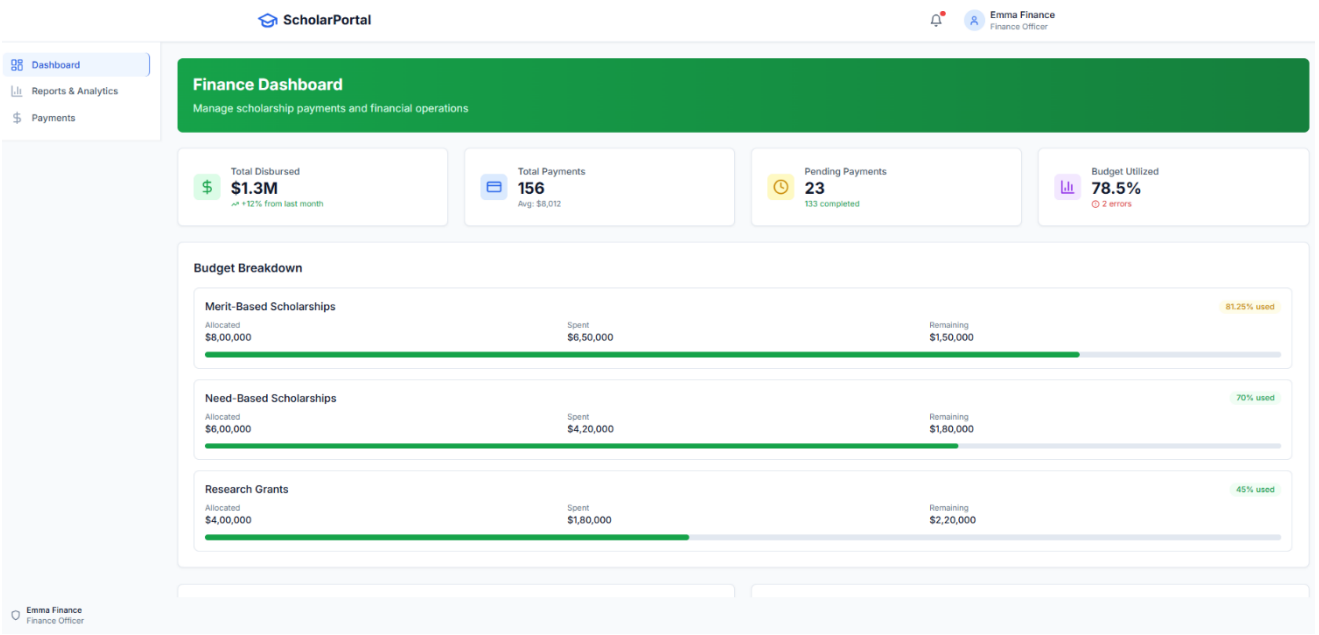


Figure 5.4: Finance Dashboard

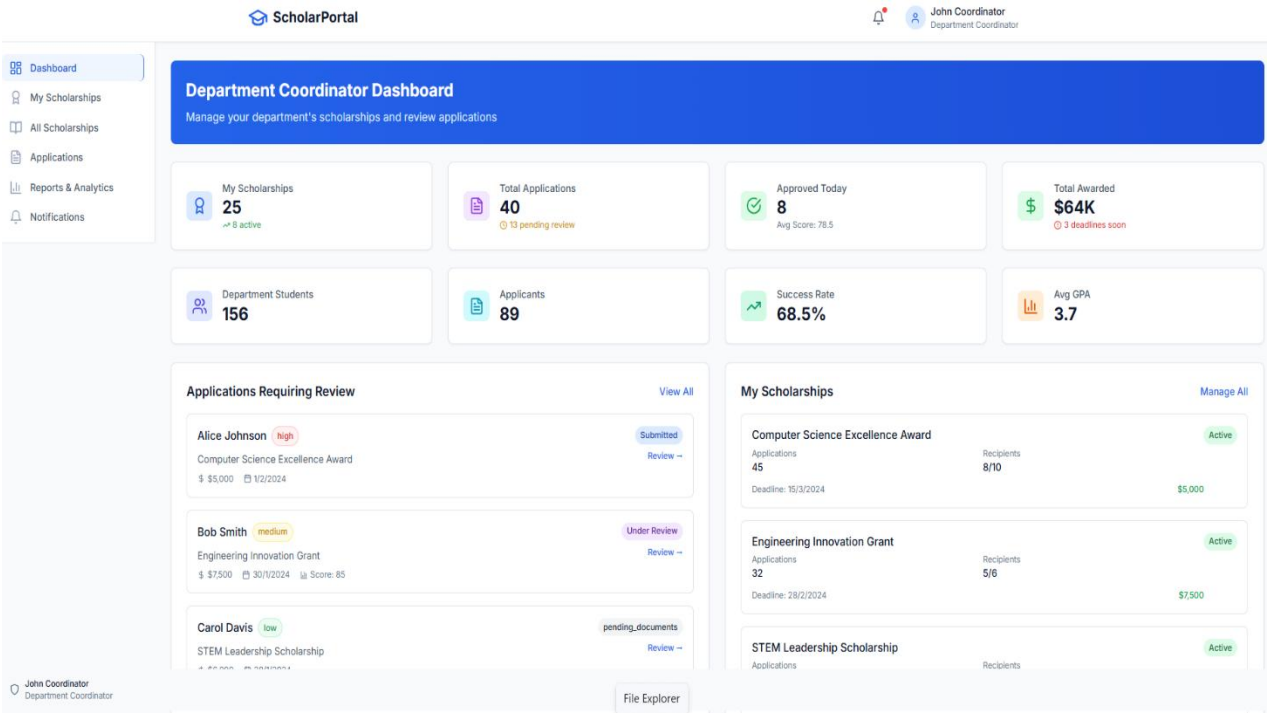


Figure 5.5: Department Coordinator Dashboard

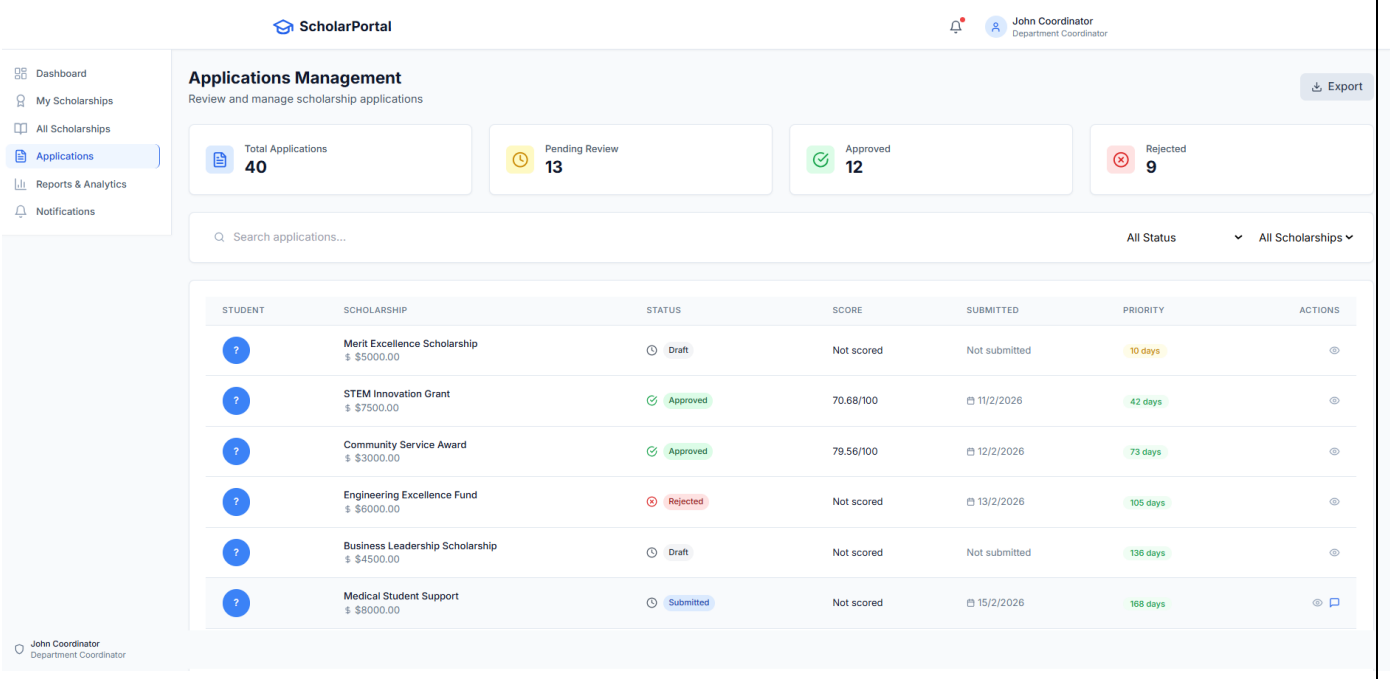


Figure 5.6: Application page

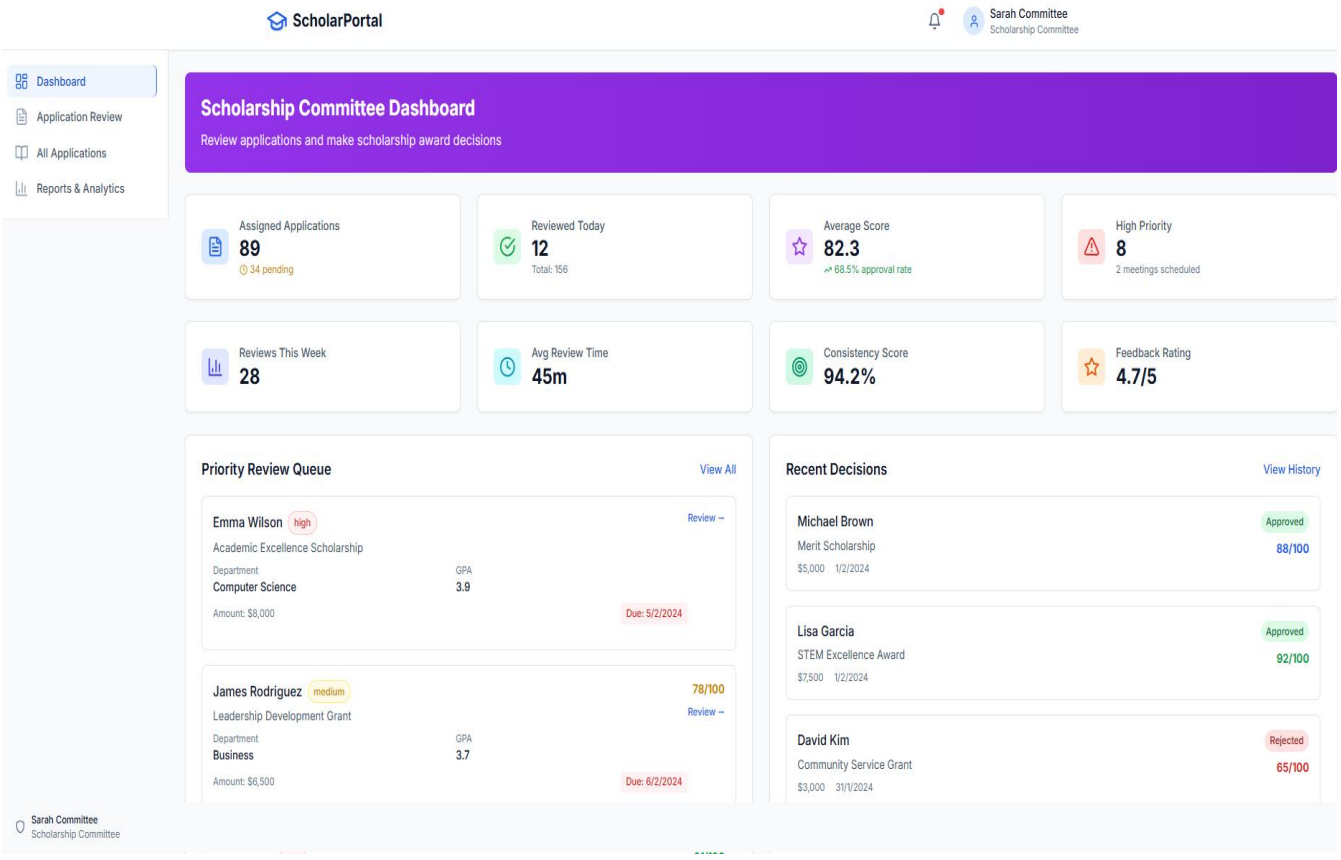


Figure 5.7: Scholarship Committee Dashboard

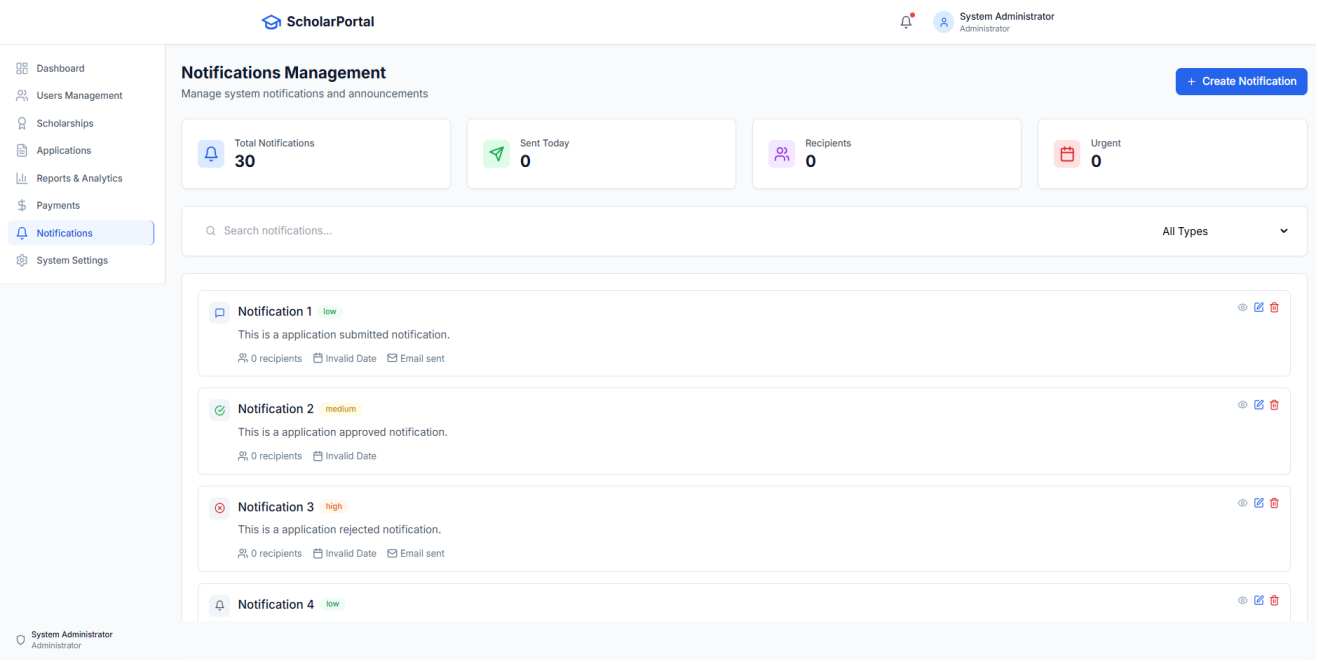


Figure 5.8: Notification page

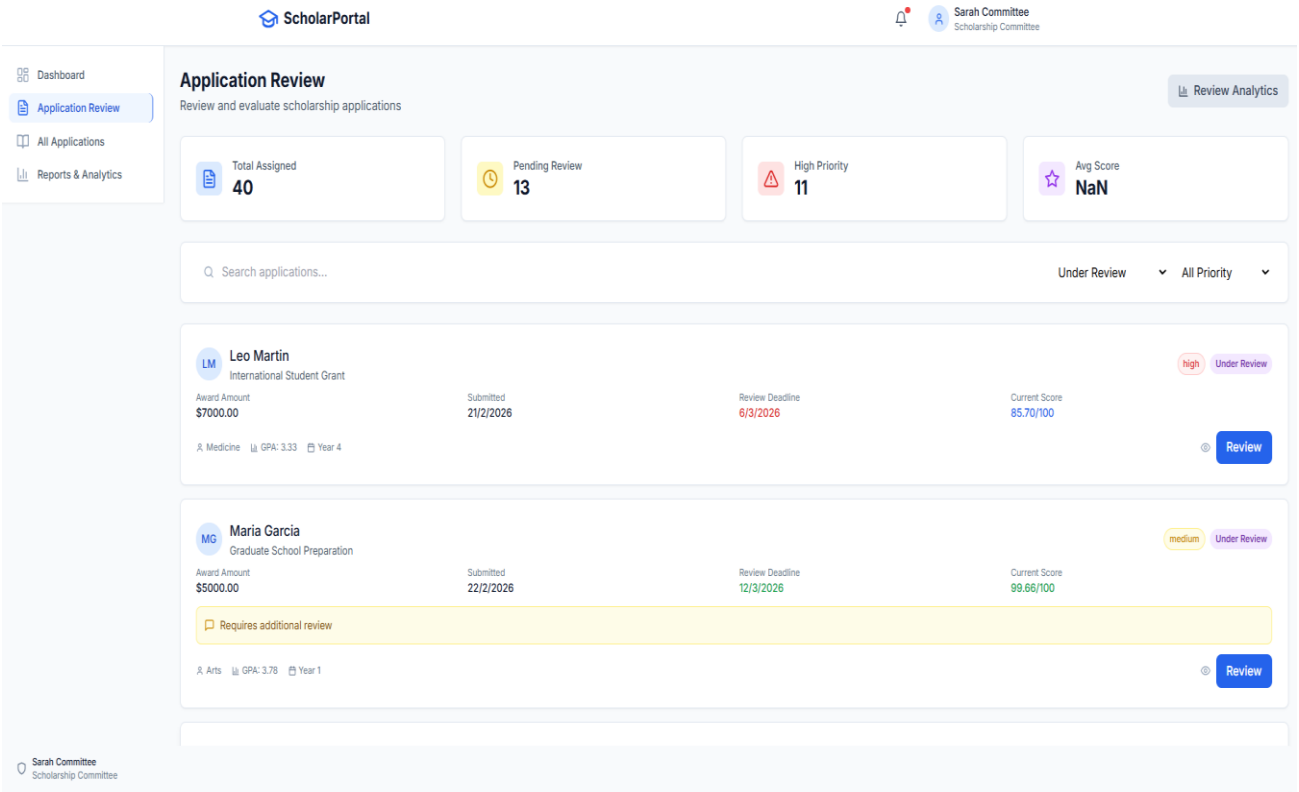


Figure 5.9: Application Review Page

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

The **Campus Scholarship and Management Portal** has been developed as an efficient web-based solution to simplify and automate the scholarship administration process within educational institutions. The system ensures transparency and accountability in the scholarship distribution process by implementing role-based access for different users such as students, department coordinators, scholarship committee members, finance officers, and administrators. Each user role has specific responsibilities in the workflow, which helps maintain proper verification and approval processes. The use of modern web technologies such as **React.js for frontend development, Node.js with Express.js for backend services, and PostgreSQL/MySQL for database management** ensures a scalable, secure, and responsive application.

Security and data integrity are maintained through features such as **JWT-based authentication, role-based access control, and secure file uploads** for student documents. The integration of notification services through email alerts helps keep students informed about application status, deadlines, and scholarship updates. The portal also maintains a structured database of student records, scholarship details, applications, approvals, and payments, which improves record management and reduces administrative workload.

Overall, the Campus Scholarship and Management Portal significantly improves efficiency, reduces manual errors, and speeds up the scholarship processing cycle. It provides a transparent and organized platform for managing scholarships, ensuring that deserving students receive financial support in a timely and systematic manner. The system demonstrates how modern web technologies can be effectively utilized to digitize institutional processes and enhance the overall management of scholarship programs within educational institutions.

6.2 FUTURE SCOPE

The Campus Scholarship and Management Portal has been successfully developed to automate scholarship management within an educational institution. However, there are several opportunities for further improvements and enhancements that can make the system more advanced, scalable, and user-friendly in the future.

Another improvement is the **implementation of mobile application support**. Developing a dedicated mobile application for Android and iOS platforms would allow students and administrators to access the scholarship portal anytime and anywhere. Mobile notifications could also be used to alert students about application deadlines, approval status, and required document updates.

A **document verification system using OCR (Optical Character Recognition)** can also be implemented. This feature would automatically extract information from uploaded documents such as income certificates, mark sheets, and identity proofs. Automated verification would speed up the approval process and reduce manual verification workload for administrators.

The system could also incorporate **SMS and push notification services** along with email notifications. This would ensure that students receive instant updates even if they are not actively checking their email accounts. Real-time alerts would improve communication and ensure that important announcements reach students quickly.

Finally, implementing **advanced security features** such as multi-factor authentication (MFA), encrypted document storage, and audit logging would further strengthen data protection and ensure compliance with institutional data security policies.

With these enhancements, the Campus Scholarship and Management Portal can evolve into a more intelligent, scalable, and fully integrated platform capable of supporting large-scale scholarship management across multiple institutions.

APPENDICES

SOURCE CODE

ADMIN DASHBOARD

```
import React, { useState, useEffect } from 'react';
import { Link } from 'react-router-dom';
import {
  Users,
  Award,
  FileText,
  DollarSign,
  TrendingUp,
  Clock,
  CheckCircle,
  XCircle,
  BarChart3
} from 'lucide-react';
import StatusBadge from '../../components/common/StatusBadge';
import LoadingSpinner from '../../components/common/LoadingSpinner';
import api from '../../services/api';

const AdminDashboard = () => {
  const [dashboardData, setDashboardData] = useState(null);
  const [loading, setLoading] = useState(true);

  useEffect(() => {
    // Fetch real dashboard data
    const fetchDashboardData = async () => {
      try {
        const response = await api.get('/admin/dashboard/stats');

        if (response.data.success) {
          setDashboardData({
            stats: response.data.data,
            recentApplications: [],
            topScholarships: [],
            monthlyStats: []
          });
        } else {
          // Fallback to mock data if API fails
          setDashboardData(mockData);
        }
      } catch (error) {
        console.error('Error fetching dashboard data:', error);
        // Fallback to mock data
```

```
    setDashboardData(mockData);
  } finally {
    setLoading(false);
  }
};

const mockData = {
  stats: {
    totalUsers: 1250,
    totalScholarships: 45,
    totalApplications: 3420,
    totalAwarded: 2850000,
    pendingApplications: 156,
    approvedApplications: 892,
    rejectedApplications: 234
  },
  recentApplications: [
    {
      id: 1,
      studentName: 'John Smith',
      scholarshipName: 'Merit Excellence Scholarship',
      amount: 5000,
      status: 'pending',
      submittedDate: '2024-01-25',
      department: 'Computer Science'
    },
    {
      id: 2,
      studentName: 'Sarah Johnson',
      scholarshipName: 'STEM Innovation Grant',
      amount: 7500,
      status: 'under_review',
      submittedDate: '2024-01-24',
      department: 'Engineering'
    },
    {
      id: 3,
      studentName: 'Mike Davis',
      scholarshipName: 'Community Service Award',
      amount: 2500,
      status: 'approved',
      submittedDate: '2024-01-23',
      department: 'Business'
    }
  ],
  topScholarships: [
```

```
      {
        id: 1,
        name: 'Merit Excellence Scholarship',
        applications: 245,
        awarded: 45,
        totalAmount: 225000,
        status: 'active'
      },
      {
        id: 2,
        name: 'STEM Innovation Grant',
        applications: 189,
        awarded: 32,
        totalAmount: 240000,
        status: 'active'
      },
      {
        id: 3,
        name: 'Community Service Award',
        applications: 156,
        awarded: 28,
        totalAmount: 70000,
        status: 'active'
      }
    ],
    monthlyStats: [
      { month: 'Jan', applications: 280, approved: 65 },
      { month: 'Feb', applications: 320, approved: 78 },
      { month: 'Mar', applications: 290, approved: 72 },
      { month: 'Apr', applications: 350, approved: 85 },
      { month: 'May', applications: 310, approved: 76 }
    ]
  };

  fetchDashboardData();
}, []);

if (loading) {
  return <LoadingSpinner text="Loading admin dashboard..." />;
}

const { stats, recentApplications, topScholarships, monthlyStats } = dashboardData;

return (
  <div className="space-y-6">
    {/* Header */}
```

```
    <div className="bg-gradient-to-r from-primary-600 to-primary-700 rounded-lg p-6
text-white">
      <h1 className="text-2xl font-bold mb-2">Admin Dashboard</h1>
      <p className="text-primary-100">
        Manage scholarships, applications, and system overview
      </p>
    </div>

    {/* Stats Grid */}
    <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-4 gap-6">
      <div className="card">
        <div className="flex items-center">
          <div className="p-2 bg-blue-100 rounded-lg">
            <Users className="w-6 h-6 text-blue-600" />
          </div>
          <div className="ml-4">
            <p className="text-sm font-medium text-academic-600">Total Users</p>
            <p className="text-2xl font-bold text-academic-
900">{stats.totalUsers.toLocaleString()}</p>
          </div>
        </div>
      </div>

      <div className="card">
        <div className="flex items-center">
          <div className="p-2 bg-purple-100 rounded-lg">
            <Award className="w-6 h-6 text-purple-600" />
          </div>
          <div className="ml-4">
            <p className="text-sm font-medium text-academic-600">Active
Scholarships</p>
            <p className="text-2xl font-bold text-academic-
900">{stats.totalScholarships}</p>
          </div>
        </div>
      </div>

      <div className="card">
        <div className="flex items-center">
          <div className="p-2 bg-green-100 rounded-lg">
            <FileText className="w-6 h-6 text-green-600" />
          </div>
          <div className="ml-4">
            <p className="text-sm font-medium text-academic-600">Total
Applications</p>
            <p className="text-2xl font-bold text-academic-
```

```
900">{stats.totalApplications.toLocaleString()}</p>
          </div>
        </div>
      </div>

      <div className="card">
        <div className="flex items-center">
          <div className="p-2 bg-green-100 rounded-lg">
            <DollarSign className="w-6 h-6 text-green-600" />
          </div>
          <div className="ml-4">
            <p className="text-sm font-medium text-academic-600">Total Awarded</p>
            <p className="text-2xl font-bold text-academic-900">${(stats.totalAwarded /
1000000).toFixed(1)}M</p>
          </div>
        </div>
      </div>
    </div>

    {/* Application Status Overview */}
    <div className="grid grid-cols-1 md:grid-cols-3 gap-6">
      <div className="card">
        <div className="flex items-center">
          <div className="p-2 bg-yellow-100 rounded-lg">
            <Clock className="w-6 h-6 text-yellow-600" />
          </div>
          <div className="ml-4">
            <p className="text-sm font-medium text-academic-600">Pending Review</p>
            <p className="text-2xl font-bold text-academic-
900">{stats.pendingApplications}</p>
          </div>
        </div>
      </div>

      <div className="card">
        <div className="flex items-center">
          <div className="p-2 bg-green-100 rounded-lg">
            <CheckCircle className="w-6 h-6 text-green-600" />
          </div>
          <div className="ml-4">
            <p className="text-sm font-medium text-academic-600">Approved</p>
            <p className="text-2xl font-bold text-academic-
900">{stats.approvedApplications}</p>
          </div>
        </div>
      </div>
```

```
      <div className="card">
        <div className="flex items-center">
          <div className="p-2 bg-red-100 rounded-lg">
            <XCircle className="w-6 h-6 text-red-600" />
          </div>
          <div className="ml-4">
            <p className="text-sm font-medium text-academic-600">Rejected</p>
            <p className="text-2xl font-bold text-academic-
900">{stats.rejectedApplications}</p>
          </div>
        </div>
      </div>
    </div>

    <div className="grid grid-cols-1 lg:grid-cols-2 gap-6">
      {/* Recent Applications */}
      <div className="card">
        <div className="flex items-center justify-between mb-4">
          <h2 className="text-lg font-semibold text-academic-900">Recent
Applications</h2>
          <Link to="/admin/applications" className="text-primary-600 hover:text-
primary-700 text-sm font-medium">
            View All
          </Link>
        </div>
        <div className="space-y-4">
          {recentApplications.map((application) => (
            <div key={application.id} className="border border-academic-200 rounded-lg
p-4">
              <div className="flex items-start justify-between">
                <div className="flex-1">
                  <h3 className="font-medium text-academic-
900">{application.studentName}</h3>
                  <p className="text-sm text-academic-600 mt-
1">{application.scholarshipName}</p>
                  <div className="flex items-center mt-2 space-x-4">
                    <span className="text-xs text-academic-
500">{application.department}</span>
                    <span className="text-xs text-academic-500">
                      ${application.amount.toLocaleString()}
                    </span>
                    <span className="text-xs text-academic-500">
                      {new Date(application.submittedDate).toLocaleDateString()}
                    </span>
                  </div>
```

```
            </div>
            <StatusBadge status={application.status} size="sm" />
          </div>
        </div>
      ))}
    </div>
  </div>

  {/* Top Scholarships */}
  <div className="card">
    <div className="flex items-center justify-between mb-4">
      <h2 className="text-lg font-semibold text-academic-900">Top
Scholarships</h2>
      <Link to="/admin/scholarships" className="text-primary-600 hover:text-
primary-700 text-sm font-medium">
        Manage All
      </Link>
    </div>
    <div className="space-y-4">
      {topScholarships.map((scholarship) => (
        <div key={scholarship.id} className="border border-academic-200 rounded-lg
p-4">
          <div className="flex items-start justify-between">
            <div className="flex-1">
              <h3 className="font-medium text-academic-
900">{scholarship.name}</h3>
              <div className="grid grid-cols-3 gap-4 mt-2">
                <div>
                  <p className="text-xs text-academic-500">Applications</p>
                  <p className="text-sm font-medium text-academic-
900">{scholarship.applications}</p>
                </div>
                <div>
                  <p className="text-xs text-academic-500">Awarded</p>
                  <p className="text-sm font-medium text-academic-
900">{scholarship.awarded}</p>
                </div>
                <div>
                  <p className="text-xs text-academic-500">Total Amount</p>
                  <p className="text-sm font-medium text-academic-900">
                    ${(scholarship.totalAmount / 1000).toFixed(0)}K
                  </p>
                </div>
              </div>
            </div>
            <StatusBadge status={scholarship.status} size="sm" />
```

```
            </div>
          </div>
        ))}
      </div>
    </div>
  </div>

  {/* Quick Actions */}
  <div className="card">
    <h2 className="text-lg font-semibold text-academic-900 mb-4">Quick
Actions</h2>
    <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-4 gap-4">
      <Link
        to="/admin/scholarships/create"
        className="flex items-center p-4 border border-academic-200 rounded-lg
hover:bg-academic-50 transition-colors"
      >
        <Award className="w-8 h-8 text-primary-600 mr-3" />
        <div>
          <p className="font-medium text-academic-900">Create Scholarship</p>
          <p className="text-sm text-academic-600">Add new scholarship program</p>
        </div>
      </Link>

      <Link
        to="/admin/users/create"
        className="flex items-center p-4 border border-academic-200 rounded-lg
hover:bg-academic-50 transition-colors"
      >
        <Users className="w-8 h-8 text-primary-600 mr-3" />
        <div>
          <p className="font-medium text-academic-900">Add User</p>
          <p className="text-sm text-academic-600">Create new user account</p>
        </div>
      </Link>

      <Link
        to="/admin/applications?status=pending"
        className="flex items-center p-4 border border-academic-200 rounded-lg
hover:bg-academic-50 transition-colors"
      >
        <Clock className="w-8 h-8 text-primary-600 mr-3" />
        <div>
          <p className="font-medium text-academic-900">Review Applications</p>
          <p className="text-sm text-academic-600">Process pending applications</p>
        </div>
```

```
        </Link>

        <Link
          to="/admin/reports"
          className="flex items-center p-4 border border-academic-200 rounded-lg
hover:bg-academic-50 transition-colors"
        >
          <BarChart3 className="w-8 h-8 text-primary-600 mr-3" />
          <div>
            <p className="font-medium text-academic-900">View Reports</p>
            <p className="text-sm text-academic-600">System analytics & reports</p>
          </div>
        </Link>
      </div>
    </div>
  </div>
 );
};

export default AdminDashboard;
```

STUDENT DASHBOARD

```
import React, { useState, useEffect } from 'react';
import { Link } from 'react-router-dom';
import { useAuth } from '../../context/AuthContext';
import { applicationService } from '../../services/applicationService';
import { scholarshipService } from '../../services/scholarshipService';
import { notificationService } from '../../services/notificationService';
import {
  Award,
  FileText,
  Clock,
  CheckCircle,
  AlertCircle,
  TrendingUp,
  Calendar,
  DollarSign
} from 'lucide-react';
import StatusBadge from '../../components/common/StatusBadge';
import LoadingSpinner from '../../components/common/LoadingSpinner';

const StudentDashboard = () => {
  const { user } = useAuth();
  const [dashboardData, setDashboardData] = useState(null);
  const [loading, setLoading] = useState(true);
```

```
  useEffect(() => {
    fetchDashboardData();
  }, []);

  const fetchDashboardData = async () => {
    try {
      setLoading(true);

      // Fetch all data in parallel
      const [applicationsRes, scholarshipsRes, notificationsRes] = await Promise.all([
        applicationService.getMyApplications({ limit: 5 }),
        scholarshipService.getScholarships({ status: 'active', limit: 5 }),
        notificationService.getNotifications({ limit: 5 })
      ]);

      const applications = applicationsRes.data.applications;
      const scholarships = scholarshipsRes.data.scholarships;
      const notifications = notificationsRes.data.notifications || [];

      // Calculate stats
      const stats = {
        totalApplications: applications.length,
        pendingApplications: applications.filter(app => ['submitted',
'under_review'].includes(app.status)).length,
        approvedApplications: applications.filter(app => app.status === 'approved').length,
        rejectedApplications: applications.filter(app => app.status === 'rejected').length,
        totalAwarded: applications
          .filter(app => app.status === 'approved')
          .reduce((sum, app) => sum + (app.scholarship?.amount || 0), 0)
      };

      setDashboardData({
        stats,
        recentApplications: applications.slice(0, 3),
        availableScholarships: scholarships.slice(0, 3),
        notifications: notifications.slice(0, 3)
      });
    } catch (error) {
      console.error('Error fetching dashboard data:', error);
      // Set empty data on error
      setDashboardData({
        stats: {
          totalApplications: 0,
          pendingApplications: 0,
          approvedApplications: 0,
```

```
        rejectedApplications: 0,
        totalAwarded: 0
      },
      recentApplications: [],
      availableScholarships: [],
      notifications: []
    });
  } finally {
    setLoading(false);
  }
};

if (loading) {
  return <LoadingSpinner text="Loading dashboard..." />;
}

const { stats, recentApplications, availableScholarships, notifications } =
dashboardData;

return (
  <div className="space-y-6">
    {/* Welcome Section */}
    <div className="bg-gradient-to-r from-primary-600 to-primary-700 rounded-lg p-6
text-white">
      <h1 className="text-2xl font-bold mb-2">
        Welcome back, {user?.first_name}!
      </h1>
      <p className="text-primary-100">
        Track your scholarship applications and discover new opportunities
      </p>
    </div>

    {/* Stats Cards */}
    <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-4 gap-6">
      <div className="card">
        <div className="flex items-center">
          <div className="p-2 bg-blue-100 rounded-lg">
            <FileText className="w-6 h-6 text-blue-600" />
          </div>
          <div className="ml-4">
            <p className="text-sm font-medium text-academic-600">Total
Applications</p>
            <p className="text-2xl font-bold text-academic-
900">{stats.totalApplications}</p>
          </div>
        </div>
```

```
      </div>

      <div className="card">
        <div className="flex items-center">
          <div className="p-2 bg-yellow-100 rounded-lg">
            <Clock className="w-6 h-6 text-yellow-600" />
          </div>
          <div className="ml-4">
            <p className="text-sm font-medium text-academic-600">Pending
Review</p>
            <p className="text-2xl font-bold text-academic-
900">{stats.pendingApplications}</p>
          </div>
        </div>
      </div>

      <div className="card">
        <div className="flex items-center">
          <div className="p-2 bg-green-100 rounded-lg">
            <CheckCircle className="w-6 h-6 text-green-600" />
          </div>
          <div className="ml-4">
            <p className="text-sm font-medium text-academic-600">Approved</p>
            <p className="text-2xl font-bold text-academic-
900">{stats.approvedApplications}</p>
          </div>
        </div>
      </div>

      <div className="card">
        <div className="flex items-center">
          <div className="p-2 bg-green-100 rounded-lg">
            <DollarSign className="w-6 h-6 text-green-600" />
          </div>
          <div className="ml-4">
            <p className="text-sm font-medium text-academic-600">Total Awarded</p>
            <p className="text-2xl font-bold text-academic-
900">${stats.totalAwarded.toLocaleString()}</p>
          </div>
        </div>
      </div>
    </div>

    <div className="grid grid-cols-1 lg:grid-cols-2 gap-6">
      {/* Recent Applications */}
      <div className="card">
```

```
        <div className="flex items-center justify-between mb-4">
          <h2 className="text-lg font-semibold text-academic-900">Recent
Applications</h2>
          <Link to="/applications" className="text-primary-600 hover:text-primary-700
text-sm font-medium">
            View All
          </Link>
        </div>
        <div className="space-y-4">
          {recentApplications.length > 0 ? (
            recentApplications.map((application) => (
              <div key={application.id} className="border border-academic-200 rounded-
lg p-4">
                <div className="flex items-start justify-between">
                  <div className="flex-1">
                    <h3 className="font-medium text-academic-
900">{application.scholarship?.name}</h3>
                    <p className="text-sm text-academic-600 mt-1">
                      Amount: ${application.scholarship?.amount?.toLocaleString() || 'N/A'}
                    </p>
                    <div className="flex items-center mt-2 space-x-4">
                      <span className="text-xs text-academic-500">
                        Applied: {new Date(application.created_at).toLocaleDateString()}
                      </span>
                      {application.submitted_at && (
                        <span className="text-xs text-academic-500">
                          Submitted: {new
Date(application.submitted_at).toLocaleDateString()}
                        </span>
                      )}
                    </div>
                  </div>
                  <StatusBadge status={application.status} size="sm" />
                </div>
              </div>
            ))
          ) : (
            <div className="text-center py-8">
              <FileText className="w-12 h-12 text-academic-300 mx-auto mb-3" />
              <p className="text-academic-600">No applications yet</p>
              <Link to="/scholarships" className="text-primary-600 hover:text-primary-
700 text-sm font-medium">
                Browse Scholarships
              </Link>
            </div>
          )}
```

```
        </div>
      </div>

      {/* Available Scholarships */}
      <div className="card">
        <div className="flex items-center justify-between mb-4">
          <h2 className="text-lg font-semibold text-academic-900">Available
Scholarships</h2>
          <Link to="/scholarships" className="text-primary-600 hover:text-primary-700
text-sm font-medium">
            Browse All
          </Link>
        </div>
        <div className="space-y-4">
          {availableScholarships.length > 0 ? (
            availableScholarships.map((scholarship) => (
              <div key={scholarship.id} className="border border-academic-200 rounded-
lg p-4">
                <div className="flex items-start justify-between">
                  <div className="flex-1">
                    <h3 className="font-medium text-academic-
900">{scholarship.name}</h3>
                    <p className="text-sm text-academic-600 mt-1">
                      Amount: ${scholarship.amount.toLocaleString()}
                    </p>
                    <div className="flex items-center mt-2 space-x-4">
                      <span className="text-xs text-academic-500 flex items-center">
                        <Calendar className="w-3 h-3 mr-1" />
                        Deadline: {new
Date(scholarship.application_deadline).toLocaleDateString()}
                      </span>
                      <span className="text-xs text-academic-500">
                        {scholarship.max_recipients - scholarship.current_recipients} spots left
                      </span>
                    </div>
                  </div>
                  <Link
                    to={`/scholarships/${scholarship.id}`}
                    className="btn-primary text-xs px-3 py-1"
                  >
                    View
                  </Link>
                </div>
              </div>
            ))
          ) : (
```

```
            <div className="text-center py-8">
              <Award className="w-12 h-12 text-academic-300 mx-auto mb-3" />
              <p className="text-academic-600">No scholarships available</p>
            </div>
          )}
        </div>
      </div>
    </div>

    {/* Notifications */}
    <div className="card">
      <h2 className="text-lg font-semibold text-academic-900 mb-4">Recent
Notifications</h2>
      <div className="space-y-3">
        {notifications.length > 0 ? (
          notifications.map((notification) => (
            <div key={notification.id} className="flex items-start space-x-3 p-3 bg-
academic-50 rounded-lg">
              <div className="flex-shrink-0">
                {notification.type === 'application_approved' ? (
                  <CheckCircle className="w-5 h-5 text-green-500" />
                ) : (
                  <AlertCircle className="w-5 h-5 text-yellow-500" />
                )}
              </div>
              <div className="flex-1">
                <p className="text-sm text-academic-900">{notification.message}</p>
                <p className="text-xs text-academic-500 mt-1">
                  {new Date(notification.created_at).toLocaleDateString()}
                </p>
              </div>
            </div>
          ))
        ) : (
          <div className="text-center py-8">
            <AlertCircle className="w-12 h-12 text-academic-300 mx-auto mb-3" />
            <p className="text-academic-600">No notifications</p>
          </div>
        )}
      </div>
    </div>
  </div>
  );
};

export default StudentDashboard;
```

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